Connected Component Based ROI Selection to Improve Identification of Microcalcification from Mammogram Images

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***Abstract*—In this paper, we propose automated image processing method for mammogram image analysis to enhance the classification of malignant and benign clustered micro calcifications. Mini-Mias, one of the most renowned mammogram dataset, is used for our experiment. Region of interest (ROI) is selected using connected component labeling method. Shape and textural features are computed and used in five different classifiers to validate the correctness of labeling method usage for this specific research. Experimental results show that Random Forest and Bagging classifier can produce best classification accuracy among them. We used Receiver operating characteristic (ROC) analysis to asses and differentiate the classification performance from different methods. Random Forest and Bagging, both classifiers provide more than 99% accuracy.**

***Index Terms*—Connected Component Labeling, Breast Cancer Detection, Mammogram Analysis, Image Processing**

I. Introduction

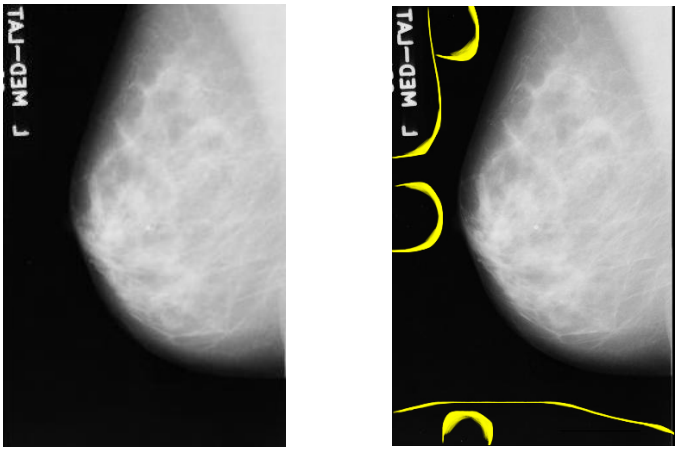
Breast cancer is one of the prevalent diseases observed among the women and the incident rate is in upward trend in recent years. There are several ways to identify breast cancer using the images from the radiologists. According to the physicians and past researchers, mammography is the most effective technique to detect this cancer at early stage. There are some research on this issue focusing on CAD (Computer Aided Diagnosis), mammogram, ultrasound and MRI (Magnetic Resonance Imaging) and clinical prototypes are built as well to improve the diagnostic performance. In this paper, we extract region of interest from Mini-Mias database and compute several features to use as feature vector for classifiers. In our approach, we apply connected component labeling to find the ROI from each mammogram and construct a feature extractor to collect features from them.

The following is a brief analysis of other contemporary research on classification of micro calcification from mammograms. Solka et al. [1] emphasized on the high dimensional features through two techniques, namely, dimensionality reduction and discriminant analysis. They used different version of contrast, smoothness, edge strength, gradient, moment, compactness, shape, and average grey level moment as features for the breast images. Later they performed Fisher Linear Discriminant (FLD) and Principal Component Analysis (PCA) to reduce dimensionality and followed Bayesian approach for classification. They show that FLD performs better than PCA while classification. Hung et al. [2] mentioned neural network to diagnose breast cancer in more depth using resampling techniques like bootstrap.They focus on building an integrative approach to build neural network models to resolve the issues of model selection, feature selection and function approximation. Their goal was to produce more accurate classification for posterior probabilities. Giger et al. [3] explained the necessity of the CAD technologies for determining the cancer indication. She mentioned three techniques, MRI, Mammography and Ultrasound and explained the comparison among them in brief. Their observation concludes that Mammogram data analysis provides better indication about the cancer detection method. Haddadnia et al. [4] used both statistical and thermal features to diagnose the breast cancer from medical infrared images. They calculated mean, variance, skewness, kurtosis and entropy to form a combinatorial model with the help of artificial neural network and genetic algorithm. It was mentioned that thermal pattern and kurtosis are the main parameters for diagnosing breast cancer form medical imaging.

Hassanien et al. [5] showed various machine intelligence techniques to diagnose breast cancer from medical images. They used type-II fuzzy algorithm to improve the quality of MRI images and then performed segmentation using pulse coupled neural networks to extract region of interests. Finally, wavelet transformation was performed on those regions to get the original information to process via support vector machine and decide whether they represent cancer or not. Huang et al. [6] used vessel analysis in 3D power doppler ultrasound images of 113 solid breast masses. Vessel analysis estimates the difference between blood vessels of tumor and non tumor areas. Six features including volume of vessels, vascularity index in normal tissue, and vascularity index in surrounding region of the tumor within 2 cm were measured; later neural network was applied to classify the tumors. The receiver operating characteristic (ROC) curve analysis was used to evaluate the performance. Lee et al. [7] proposed an efficient classifier with feature selection based on fuzzy entropy for pattern classification which was formulated from Shanon entropy.They performed their experiment in Iris and Wisconsin Breast Cancer Diagnostic Database(WBCDB).

The rest of the paper is organized as follows: A brief

discussion on region of interest (ROI) selection method, mathematical formulation, and description on connected component technique and its application in our research is presented in section II. Performance evaluation and results are discussed in section IV. Finally, section V concludes and suggests future direction of our research.



(a) (b)

Fig. 1. (a) Raw X-ray mammogram from MiniMias Dataset (image number mdb029.pgm). (b) Generalized marking of the objects detected by the connected component labeling algorithm.

II. ROI Selection Using Connected Component

In this paper, the mammogram image [8] classification is separated in two groups with seven classes, i.e., an image can be either clustered with micro-calcification or not. These two classes are known as malignant and benign by the radiologists. It is not obvious that a breast tissue with malignant micro-calcification is sure to have some cancer in it.

The first step is to preprocess the mammogram images which is the raw image data from the X-ray machine. The image contains additional data(mammogram date, difference in background and foreground due to the imbalance of the light rays, shade, spots etc.) that can be put aside while detecting the lesion grand. But the question comes how computers can ignore these noise? To begin, let vector $x_i^j \subset R^N$ denote the set of points for different objects within an image. In addition, i and j refers number of objects and pixel values within each objects those are not null and have distance less or equal to some predefined threshold.

$$x_i^j = \{1, \ldots, j\} \tag{1}$$

Most of the previous work was performed with the help of image cropping using tools like Adobe Photoshop that can invalidate useful pixel values while conversion from one format to another. Thus, we avoid using those methods to resize the image. In general, it is evident from the mammogram images that the largest area covered must preserve the breast gland. So, our aim is to look for the available objects within the image first, which makes the problem into an object detection problem. If we look into Fig.1 (b), it is noticeable that the yellow marks are showing different smaller objects within the image area. White letters within the textual part are treated individually while leaving them out in the preprocessing step, but it is shown in a group for the simplicity. Additionally, the images are considered to be black and white for this experiment.

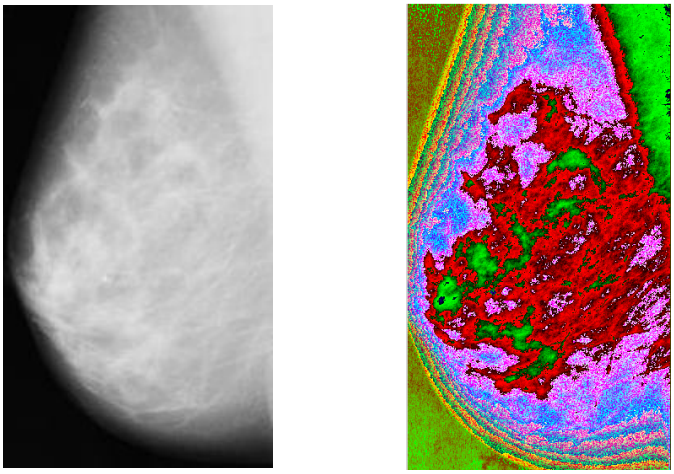
Mammograms are taken from different angles to ensure that no area from the tissue is being abandoned, later the X-ray film is examined by specialists for further verification. Now, our aim is to simulate the specialists verification knowledge through CAD (Computer Aided Diagnosis) systems.

Before going in detail, first we should know what connected sets determine? Mathematically, A region $R \subset S$ is defined as connected under $c(s)$ if for all $s, r \in R$, there exists a sequence of M pixels, $s_1, \ldots, s_M$ such that $s_1 \in c(s), s_2 \in c(s_1), \ldots, s_M \in c(s_{M-1}), r \in c(s_M)$. Which mean there must be a connected path from s to r. Now, as we are concerned about grayscale image which has values between 0 to 255 in each pixel, we can start from any pixel which has value 0 (black pixel treated as background) and start to grow each connected set. Additionally, we chose the neighborhood system to select the nearest pixels for region growing purpose. Usually, there are two neighborhood systems, i.e., 4-point neighborhood and 8-point neighborhood method denoted as N_4 and N_8 respectively.[9]

	N	
W	S_0	E
	S	

NW	N	NE
W	S_1	E
SW	S	SE

Fig. 2. Demonstration of neighborhoods of a pixel. S_0 is at the center when 4-point neighborhood is considered. S_1 is the central pixel surrounded by eight neighboring pixels. The four neighborhood of $N_4(x, y)$ of pixel (x, y) is surrounded with pixels $(x-1, y), (x+1, y), (x, y+1), (x, y-1)$. Similarly, $N_8(x, y)$ of pixel (x, y) includes with neighboring pixels of $(x-1, y), (x-1, y+1), (x-1, y-1), (x+1, y), (x+1, y+1), (x+1, y-1), (x, y+1), (x, y-1)$. N,E,W,S and other four derived values denotes the side of the center pixel.



(a) (b)

Fig. 3. (a) Cropped image without removing any pixel information from the region of interest. (b) Colormap applied for better visualization of abnormal tissue image in the ROI.

8-points neighborhood pixel method is suitable for use in RGB images where objects have different combination of colors and the pixels values vary too much in general. On the contrary, gray scale images can have only values between 0 to 255 and most dominant colors are black and white, thus the difference between white and black sectors are very much prevalent to distinguish. As a result 4-point neighborhood method works well to determine the connected objects with

ease, but there is some relation with the threshold value which has to be checked each time the region is grown [10] from the source S_0 (Fig.2). Now we can formulate that the connected set, $c(s) = \{r \in \delta_s : |S_r - P_s| < Th\}$, where, r is the pixel position to be tested, δ_s is the set of neighboring pixels, $S_r - P_s$ refers to the distance between pixel r and each neighbors, and Th is the threshold value that can not be crossed. However, two or more objects may overlap in some scenarios, which can be difficult to differentiate for gray scale images compared to RGB ones. Connected component algorithm can well identify the presence of the overlapping objects by computing the intensities of the color level, but it is not easy in our case as the black color is considered to be the background and the only important values for pixels are those which are non zero values.

III. Description of Mammogram Dataset and Feature Selection

A. Mammogram Data Set

We performed our experiment using a dataset prepared by the Department of Physics at the Royal Marsden Hospital of London, UK and it is popular by its name Mini-Mias mammogram repository [11]. The data contains 322 images of 161 patients, each one having left and right mammograms. All the images size were adjusted by padding zeroes (black background) to make 1024×1024 pixels. Each image is accompanied with some predefined information, i.e., center of the image, radius, coordinate start location etc. It was also mentioned whether calcification is present centrally or in a scattered manner. Normal and benign classes are mentioned along with other six abnormality types. The origin of the abnormality is provided with their axis values and the radius of the calcification spread. For example, the calcification origin is at (500,200) and the radius is given as 312, which means the abnormality is approximately spread around the point (500,200) with the radius of 312 pixel units.

B. Feature Extraction and Processing

For automated classification, we used total ten features of both shape and texture category and saved in a vector. For example, if an image, A has radius, r and center, c as features, then the feature vector of image A is $V(A) =< r, c >$. The features extracted from the processed mammograms are variance, standard deviation, median, mode, average gray scale value, average contrast, smoothness, skewness, uniformity, and entropy. The first four features are statistical measures and the latter six are texture features. Average gray-scale value is computed from the first moment which gives the idea about the distribution of the gray-level pixels. Average contrast is measured by the standard deviation σ, which is the square root of the second moment. Smoothness [12] provides the approximating function to capture significant patterns from the pixel data by removing the noise. We used the below equation for smoothing purpose.

$$R = 1 - \frac{1}{1 + \sigma^2} \tag{2}$$

Skewness can be positive, negative or both to quantify the bias of the random variable (here pixel) from the ideal viewpoint.[13] It is measured using the third moment. Equation 3 shows the correspondence among third central moment (μ_3), mean (μ), standard deviation (σ), and the expectation operator (E).

$$\gamma_1 = E\left[\left(\frac{X - \mu}{\sigma}\right)^3\right] = \frac{\mu_3}{\sigma^3} = \frac{E[(X - \mu)^3]}{(E[(X - \mu)^2])^{3/2}} \tag{3}$$

Uniformity of the mammogram measures the sum of the squared relative frequencies p_i of the gray-scale values and it is maximum when image remain constant. Entropy E measures the predictability using Equation 4.

$$E = p_i \log_2 p_i \tag{4}$$

We masked [14] the original mammograms after processing them using connected component algorithm in order to compute these features for further classification. As discussed in the next section, several classifiers were run to identify the classification accuracy parameters using the feature vector.

IV. Evaluation

Statistical and texture features were extracted from the Mini-Mias dataset using Matlab programs. Microcalcification detection was performed using decision tree, bagging, decision table, and locally weighted learning classifier available in the Weka Machine Learning Algorithms Toolkit [15]. This experiment was evaluated using 10-fold cross validation for the classifiers to split the input feature vector into both training and testing set.

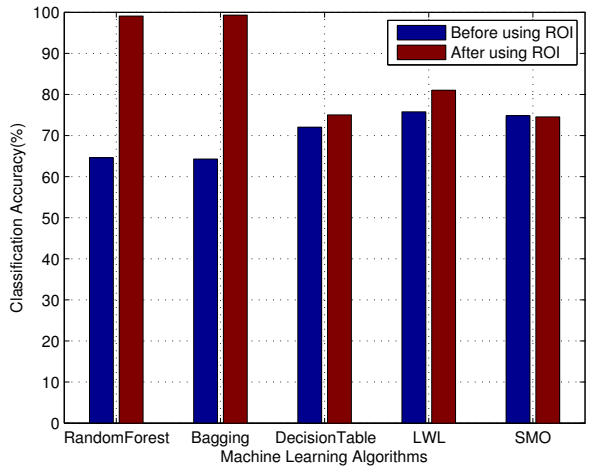


Fig. 4. Classification accuracy and type of the algorithms used with the data vectors extracted from the processed mammograms. Random Forest and Bagging provides similar accuracy both before and after the usage of labeling.

A. Results and Discussion

Classifiers were tested with several configuration using the dataset containing 322 samples from the MiniMias. Experimental result shows that Bagging [16] and Random Forest [17] classifier provides the maximum accuracy (Fig. 4) among the chosen five algorithms. Random Forest is of type decision tree classifier, which uses multiple decision trees at the training time and the mode class among them becomes the desired output class (99.06% accuracy). Random forest of 10 trees, each constructed while considering 5 random features and the out of bag error was 0.3571. Bagging was used with 100% bag size, J48 pruned tree, and seed value of 1.00. It provides 99.37%

TABLE I. TP (True Positive), FP (False Positive) and ROC (Receiver Operating Curve) values shows strong class label detection. (Sample shown for Bagging classifier)

TP Rate	FP Rate	ROC Area	Class
1.00	0.00	1.00	CIRC
1.00	0.017	1.00	NORM
0.929	0.00	1.00	MISC
1.00	0.00	1.00	ASYM
1.00	0.00	1.00	ARCH
1.00	0.00	1.00	SPIC
0.96	0.00	1.00	CALC
0.994	0.011	1.00	Weighted Avg.

accuracy for the classification. Particularly, Bagging classifier reduces the variance among the output class using sampling with replacement. Locally weighted learning (LWL) algorithm used decision stump as classifier, Linear nearest neighbor as neighbor search algorithm, and the value for KNN weighted function as 1 to select all the neighboring values (80.43% accuracy). Decision table majority classifier used leave-one-out cross validation method, forward best first search, and RMSE (Root mean squared error) as evaluation measure. SMO (Sequential Minimal Optimization) used Polykernel as kernel function along with 0.001 as tolerance parameter and provided 74.53% accuracy.

Table I shows the ROC (Receiver operating characteristics) [18], TP (True positive), and FP (False positive) values for each of the classes for Bagging classifier. The weighted ROC value is 1.00, which means the maximum accurate performance is achieved with this classification. TP rate and FP rate helps perceive the misclassification error rate of the classifier. For example, let there are two classes of A, B and TP rate of 0.8 and 0.9 respectively (FP rate is 0.2 and 0.1 as it is the complement of TP rate). So, it means 80% of class A data is picked as class A and the rest 20% is misclassified as class B. Thus, higher value of TP rate for a particular class defines higher classification accuracy. (TP and FP rate ranges between 0 and 1)

On the other hand, we can notice significant improvement of classification accuracy for both Random Forest and Bagging classifier when compared with non-connected component labelled mammograms. One of the major reason for this improvement is normalizing the frequency distribution of the pixels, which is ensured by the labeling operation. Additionally, all the four classifier provides better result than before, except SMO which shows a bit less. SMO uses Support vector machine (SVM), which is best known for classification of noisy data of two class. In our experiment, six classes were present and there were some overlapping of feaures of different classes that can make the classification task confusing for SVM.

Table I shows seven different class labels provided in the dataset, which can further be narrowed down to two class problem (Normal and Abnormal) [19]. These six other classes, except normal category, defines different type of malignant tissue growth within breast gland.

V. Conclusions

Using regression model and decision tree can enhance the classification accuracy in great detail, compared to other algorithms for this specific mammogram classification research. However, our major goal was to augment the mammography image processing to collect the desired region from them, which in the end helps exposing the abnormality or severity of breast cancer via computer aided applications. Thus, we had to go through the classification process as well to prove that the detection accuracy improves if our ROI selection method is followed. This experiment was performed using 10-fold cross validation with labeled dataset.We will use larger dataset and narrow down the seven class problem into two class classification problem using layered architecture in our future research. The first layer will separate the seven classes. The second class will leave *Normal* category intact and run the classification on the rest six classes. We will use our method on separate testing and training data as well, which can surely help diagnose early stage cancer identification.

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